

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Scala For Data Science

Practical IV

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Class: SYBsc IT	Subject: Scala For Data Science
Date of Assignment: 14/07/2026	Date/Time of Submission: 14/07/2026

AIM: Generate a random dataset of 10 numbers and calculate its variance and standard deviation.

CODE:

```
build.sbt  IQRA.scala x
1  import scala.util.Random
2  object Demo extends App{
3      val data = Array.fill(10)(Random.nextInt(100))
4      println("Random Dataset: "+ data.mkString(", "))
5      val mean = data.sum / data.length
6      val s = data.map(x => math.pow(x - mean,2))
7      val v = s.sum / data.length
8      val d = math.sqrt(v)
9
10     println(f"Mean: $mean")
11     println(f"Variance: $v%.2f")
12     println(f"Standard Deviation: $d%.2f")
13     println("IQRA S021")
14
15
16
17 }
```

IF LENGTH IS 20 EXAMPLE:

```
build.sbt  IQRA.scala x
1  import scala.util.Random
2  object Demo extends App{
3      val data = Array.fill(20)(Random.nextInt(100))
4      println("Random Dataset: " + data.mkString(", "))
5      val mean = data.sum / data.length
6      val s = data.map(x => math.pow(x - mean,2))
7      val v = s.sum / data.length
8      val d = math.sqrt(v)
9
10     println(f"Mean: $mean")
11     println(f"Variance: $v%.2f")
12     println(f"Standard Deviation: $d%.2f")
13     println("IQRA S021")
14
15
16
17 }
18
```

OUTPUT:

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
Random Dataset: 65, 84, 50, 3, 6, 88, 99, 18, 77, 21
Mean: 51
Variance: 1195.30
Standard Deviation: 34.57
IQRA S021

Process finished with exit code 0
```

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
Random Dataset: 99, 23, 71, 91, 37, 49, 97, 76, 61, 21, 38, 76, 94, 53, 0, 41, 79, 71, 35, 45
Mean: 57
Variance: 746.45
Standard Deviation: 27.32
IQRA S021

Process finished with exit code 0
```